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# Weather-Conditioned Deep Learning for Bird Migration Trajectory Forecasting

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## Abstract

Forecasting bird migration from GPS traces is challenging because most daily movements are small or stationary, while some days contain long directed flights. We study next-day southbound bird trajectory prediction using recent GPS history and optional ERA5 weather features. Our experiments compare direct coordinate regression with a triline formulation that predicts fly/no-fly, distance, and heading before reconstructing the next location. We also compare MLP, LSTM, and Transformer-based models under context lengths  $k \in \{7, 14, 30\}$ . The Triline Transformer 4L gives the best overall next-day performance among the main models, reaching 15.76 km mean error and 2.31 km median error without weather at  $k = 30$ . Longer context generally helps the Transformer, while weather features provide mixed benefits and do not consistently reduce mean error. Autoregressive rollout results show that one-step errors can accumulate over full migration paths, making long-horizon forecasting more difficult.

## 1 Introduction

Bird migration prediction is a spatiotemporal forecasting problem. Given several days of GPS observations from an individual bird, our goal is to predict the bird’s next-day latitude and longitude. This task is difficult because migration trajectories contain both stopover periods and long flights, so the model needs to learn not only where the bird is, but also whether it is likely to move and how far it moves.

In this project, we study southbound migration of White Stork (*Ciconia ciconia*) using daily GPS trajectories and aligned ERA5 weather features. We compare two prediction styles. The first directly predicts latitude and longitude displacement. The second uses a triline movement formulation, where the model predicts fly/no-fly, movement distance, and heading before converting them back into GPS coordinates. This structure better matches the movement behavior in migration data.

We evaluate several models, including a Direct MLP baseline, a Triline LSTM, and a Triline Transformer. We also test different history lengths,  $k = 7, 14$ , and  $30$ , to see whether longer trajectory context helps the model understand the migration phase. Finally, we compare models with and without weather features to study whether environmental information improves prediction.

Our results show that the Triline Transformer 4L performs best overall in the main next-day prediction experiments. In the no-weather setting, it achieves the lowest mean, median, and P90 errors at  $k = 30$ , suggesting that both movement decomposition and attention-based history modeling are useful. Weather features have a mixed effect: they help some settings but do not consistently improve the average error. In autoregressive rollout, errors grow much larger than one-step errors, showing that full-path migration forecasting remains challenging.

## 2 Related Work

Bird migration forecasting and animal movement modeling connect ecological tracking, weather-driven migration studies, statistical movement models, and modern sequence learning. Large-scale tracking platforms such as Movebank make individual-level GPS trajectory analysis possible by standardizing animal movement records across species and studies (1). Ecological studies have also shown that atmospheric conditions, especially wind and weather, can create favorable migration routes, delays, detours, and other movement constraints (2). At larger spatial scales, BirdCast-style systems use atmospheric variables and radar observations to forecast continental migration intensity (3), while MistNet applies convolutional neural networks to extract bird migration signals from archived weather radar data (4). These works motivate weather-conditioned migration modeling, but they mainly focus on aggregate migration intensity or radar-derived movement patterns rather than future GPS trajectories of individual birds.

Traditional animal movement models provide interpretable frameworks for modeling individual movement behavior. Markov models, correlated random walks, dynamic Brownian bridge movement models, and step-selection approaches describe movement using local step dynamics, utilization distributions, or environmental constraints (5; 6). Integrated step selection analysis, for example, combines movement dynamics with habitat selection to study how animals choose their next step under environmental conditions (6). These methods are grounded in ecological theory and are useful for interpreting movement mechanisms, but they often rely on local dependencies and parametric assumptions. This limits their flexibility for modeling nonlinear interactions among recent trajectory history, behavioral state, and weather variables over longer temporal windows.

Deep learning methods provide more flexible tools for learning temporal patterns from trajectory and environmental data. LSTMs are natural baselines for movement forecasting because they process trajectory history sequentially, but they compress past observations into hidden states and may struggle with long-range dependencies. Transformer models use self-attention to compare all time steps in the context window directly, which has motivated their use in long-sequence time-series forecasting models such as Informer and Autoformer (7; 8). In animal movement modeling, MoveFormer applies a Transformer-based framework to step-selection prediction using past trajectories, temporal information, and environmental covariates (9). DeepSSF also combines ecological step-selection ideas with neural networks to learn movement and environmental selection in a data-driven way (10). These works show that neural models can extend classical movement models while retaining movement-structured prediction targets.

Our project builds on this literature but focuses on a different forecasting setting. Existing work often predicts aggregate migration intensity, estimates utilization distributions, or models local step selection, while our goal is individual-level bird migration trajectory forecasting. We compare direct GPS coordinate regression with a structured triline formulation that predicts fly/no-fly behavior, movement distance, and heading before reconstructing the next location. We also evaluate both next-day prediction and autoregressive rollout, since full trajectory forecasting requires stability when model predictions are fed back into future inputs. Finally, we test whether ERA5 weather covariates improve individual trajectory prediction. In this way, our work connects ecological movement modeling, weather-conditioned migration forecasting, and Transformer-based sequence learning for individual bird migration trajectories.

## 3 Data

### 3.1 Dataset Description

We use public Movebank LifeTrack White Stork GPS tracking data from SW Germany and Bavaria. The original records contain high-frequency GPS observations, including timestamp, latitude, longitude, and individual bird identifier. To make the task suitable for daily trajectory forecasting, we aggregate the raw GPS observations into daily path records. For each bird and date, we use a representative daily location and compute movement-related features such as latitude, longitude, daily displacement, step length, heading, day-of-year encoding, and stopover duration. The two regional datasets are combined to increase route diversity while keeping the species and tracking protocol consistent.

Weather context is obtained from ERA5 hourly single-level reanalysis data. ERA5 variables are aligned with the daily bird records by date and nearest spatial grid cell. The weather features include 2-meter temperature, 10-meter wind speed, wind direction, surface pressure, precipitation, cloud cover, and boundary layer height. Precipitation is log-transformed before modeling to reduce the influence of heavy-tailed extreme values.

### 3.2 Dataset Cleaning

We first merged the daily bird movement records with aligned weather covariates. Each GPS-based daily record was matched with the corresponding weather observation for the same bird and date when available; otherwise, the nearest same-bird weather record was used. The final input therefore combines path-level movement features with environmental context.

To focus on meaningful fall migration behavior, we applied several filtering steps. We removed birds with fewer than 60 surviving daily records, since very short trajectories provide limited historical context for sequence modeling. We then restricted the date range to August 1 through December 31, which captures the primary southbound migration season. Spatially, we kept only southbound migration paths within the latitude band from 55°N to 30°N, so the model is trained on trajectories that correspond to the main migratory movement rather than local or northbound motion.

After cleaning, the dataset contains 13,550 daily records from 135 migration paths and 71 source birds. For model evaluation, we used a chronological 80/20 split based on the target prediction date: the earlier 80% of samples were used for training and the later 20% were held out for testing. This avoids temporal leakage and better reflects the intended forecasting setting, where models predict future migration behavior from past observations.

Table 1: Dataset and split summary.

| Field               | Value                                    |
|---------------------|------------------------------------------|
| Daily records       | 13,550                                   |
| Migration paths     | 135                                      |
| Source birds        | 71                                       |
| Date range          | 2013-07-30 to 2023-12-26                 |
| Context lengths     | $k = 7, 14, 30$ days                     |
| Train/test protocol | Chronological 80/20 split by target date |
| Prediction target   | Next-day latitude and longitude          |
| Fly threshold       | 10 km daily displacement                 |

The final weather-augmented input uses 18 features per daily record. The path features include latitude, longitude, daily latitude/longitude displacement, step length, heading encoded as sine and cosine, day-of-year encoded as sine and cosine, and stopover duration. The weather features include 2-meter temperature, 10-meter wind speed, wind direction encoded as sine and cosine, surface pressure, log-transformed precipitation, cloud cover, and boundary layer height. These features are concatenated across the selected context window of  $k = 7, 14$ , or 30 days before predicting the next-day location.

## 4 Methods

### 4.1 Problem Formulation

We formulate migration forecasting as a next-day location prediction task. For each migration path, the model observes a historical window of daily feature vectors

$$\mathbf{x}_{t-k+1:t} = [\mathbf{x}_{t-k+1}, \dots, \mathbf{x}_t],$$

where  $k \in \{7, 14, 30\}$  is the context length. Each feature vector contains path features and, in the weather-augmented setting, aligned environmental features. Given the current location

$$\mathbf{s}_t = (\text{lat}_t, \text{lon}_t),$$

the goal is to predict the next-day location  $\hat{\mathbf{s}}_{t+1}$ .

Model performance is evaluated using haversine distance between the predicted next-day location and the observed next-day location. We report mean error, median error, and P90 error. Mean error summarizes overall prediction accuracy across all target days, while median error describes the typical case and is less sensitive to large outliers. P90 error measures tail performance by reporting the error threshold below which 90% of predictions fall.

## 4.2 Direct Regression Baselines

The direct regression models predict the next-day coordinate displacement:

$$\widehat{\Delta \mathbf{s}}_{t+1} = [\widehat{\Delta \text{lat}}_{t+1}, \widehat{\Delta \text{lon}}_{t+1}], \quad \widehat{\mathbf{s}}_{t+1} = \mathbf{s}_t + \widehat{\Delta \mathbf{s}}_{t+1}.$$

This approach is simple because it directly optimizes the final GPS target. However, it combines several different behaviors into one regression output: whether the bird moves, how far it moves, and which direction it moves. This can make the prediction less interpretable and may lead to overshooting during autoregressive rollout, especially when the model predicts moderate displacement for both stationary days and true migration days.

## 4.3 Triline Movement Decomposition

To make the prediction more structured, we decompose movement into three components: fly/no-fly, distance, and direction. Given an encoded history representation  $\mathbf{h}$ , the triline model predicts

$$\widehat{p}_{fly} = \sigma(f_{fly}(\mathbf{h})), \quad (1)$$

$$\widehat{d} = \exp(f_{dist}(\mathbf{h})) - 1, \quad (2)$$

$$[\widehat{\sin \theta}, \widehat{\cos \theta}] = f_{dir}(\mathbf{h}). \quad (3)$$

The fly head predicts whether the bird moves more than the 10 km daily displacement threshold. The distance head predicts log-transformed step length, and the direction head predicts heading as sine and cosine to avoid angular discontinuity.

The training objective combines the three prediction tasks:

$$\mathcal{L} = \text{BCE}(y_{fly}, \widehat{p}_{fly}) + \text{MSE}(\log(1 + d), \log(\widehat{1} + \widehat{d})) + 0.5 \text{MSE}(\mathbf{u}, \widehat{\mathbf{u}}),$$

where  $\mathbf{u} = [\sin \theta, \cos \theta]$ . The direction loss is mainly applied to moving samples because heading is not meaningful for stationary days.

At inference time, the predicted distance and direction are converted back into latitude and longitude displacement:

$$\widehat{\Delta \text{lat}} = \frac{\widehat{d} \cos(\widehat{\theta})}{111}, \quad \widehat{\Delta \text{lon}} = \frac{\widehat{d} \sin(\widehat{\theta})}{111 \cos(\text{lat}_t)}.$$

The final next-day location is reconstructed from the current location and the predicted displacement. For Triline prediction, the predicted distance is set to zero when  $\widehat{p}_{fly}$  is below the selected threshold.

## 4.4 Triline Transformer Architecture

The proposed Triline Transformer takes the past  $k$ -day trajectory window as input (Figure 1). Each daily feature vector is first projected into a hidden embedding space, then combined with positional embeddings. The sequence is passed through Transformer encoder layers with multi-head self-attention, feed-forward blocks, GELU activation, dropout, and normalization. Self-attention allows the model to compare different days in the context window and learn which historical observations are most useful for predicting the next movement.

After sequence encoding, the model applies learned attention pooling over the  $k$  encoded days to obtain a single history representation  $\mathbf{h}$ . This avoids relying only on the last timestep and allows the model to summarize information from the full trajectory window. The pooled representation is then passed to three separate prediction heads for fly/no-fly probability, movement distance, and movement direction.

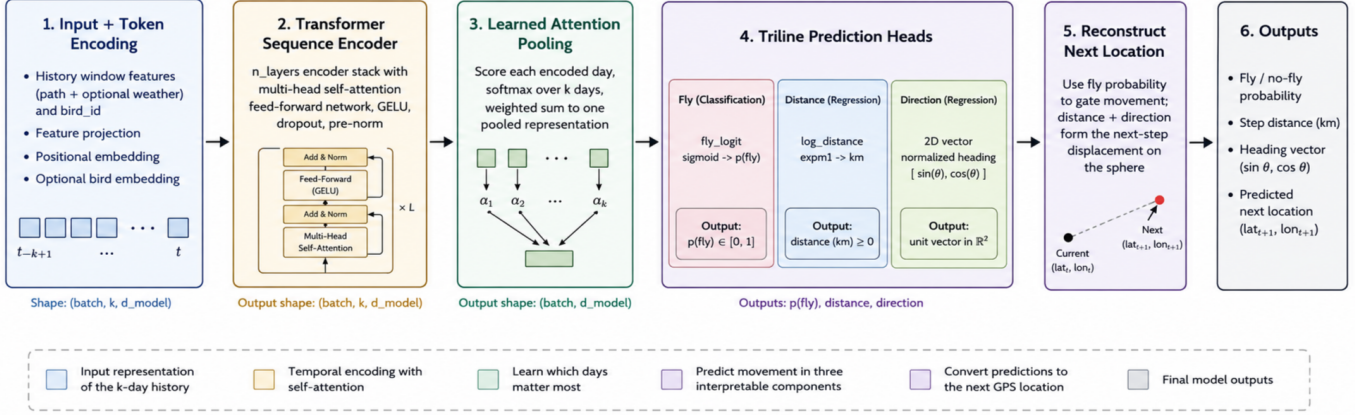


Figure 1: Triline Transformer block diagram. The model encodes a  $k$ -day movement and weather history, applies learned attention pooling, and predicts fly/no-fly probability, distance, and direction before reconstructing the next GPS location.

## 4.5 Model Comparisons

We compare three model families in the experiments. The Direct MLP flattens the  $k$ -day input window and predicts coordinate displacement using feed-forward layers. The Triline LSTM uses a recurrent encoder with the same three triline prediction heads. The Triline Transformer combines self-attention with the triline output structure.

The first comparison evaluates the core next-day GPS forecasting task. By comparing direct displacement regression with triline movement prediction, we test whether separating movement into fly/no-fly, distance, and direction provides a more behavior-aligned target than predicting latitude and longitude displacement directly. The comparison between Triline LSTM and Triline Transformer then isolates the effect of the temporal encoder, since both models use the same triline output heads but encode the history differently.

We also vary the context length with  $k = 7, 14$ , and 30 days of past observations. These values represent short, intermediate, and approximately one-month trajectory histories, allowing us to test whether next-day migration prediction depends mainly on recent local motion or on longer migration-phase context. The MLP baseline provides a non-sequential reference point, while the LSTM and Transformer test whether ordered temporal modeling improves the use of longer histories.

We further compare models with and without ERA5 weather covariates. This ablation tests whether environmental variables provide useful predictive signal beyond GPS-derived trajectory history.

Finally, we evaluate autoregressive rollout on held-out migration paths. Unlike next-day prediction, rollout repeatedly feeds the model’s predicted location back into the input context to forecast multiple future steps. This comparison tests whether each model remains stable over longer horizons and whether one-step errors accumulate into large trajectory-level deviations.

## 5 Experimental Setup

All experiments use seed 42, CUDA, AdamW with learning rate  $10^{-3}$  and weight decay  $10^{-4}$ , batch size 128, maximum 200 epochs, and early stopping patience 30. The learning rate scheduler is ReduceLROnPlateau with factor 0.5 and patience 4. The chronological split sorts all windows by target date, assigns the first 80% to training, and assigns the last 20% to testing without shuffling. The current training code uses this test partition for learning-rate scheduling, early stopping, and checkpoint selection.

Table 2: Model sizes in the experiment. Parameter counts vary slightly with feature mode and context length; values shown are representative for  $k = 30$ .

| Model                  | Parameters without weather | Parameters with weather |
|------------------------|----------------------------|-------------------------|
| Direct MLP             | 47K                        | 78K                     |
| Triline LSTM 4L        | 2.16M                      | 2.16M                   |
| Triline Transformer 4L | 3.30M                      | 3.30M                   |

## 6 Results

### 6.1 Next-Day Prediction

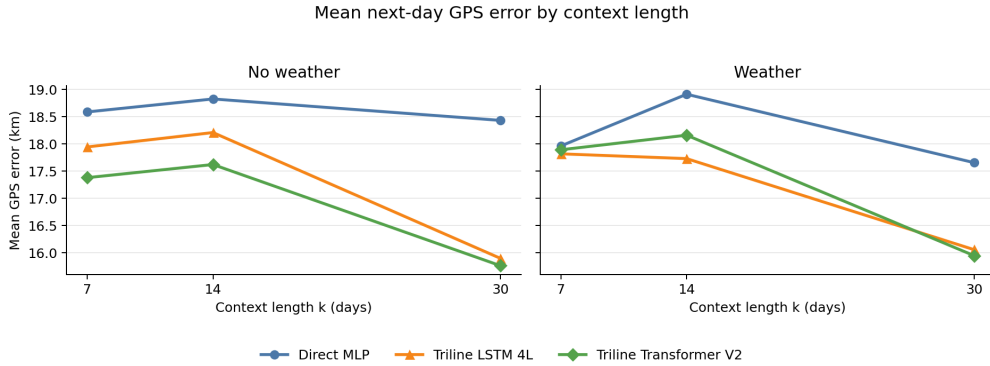


Figure 2: Mean next-day GPS error across context lengths.

Table 3: Next-day prediction results for  $k = 30$ . Lower error is better.

| Condition  | Model                         | Mean km      | Median km   | P90 km       |
|------------|-------------------------------|--------------|-------------|--------------|
| No weather | Direct MLP                    | 18.43        | 5.26        | 50.98        |
| No weather | Triline LSTM 4L               | 15.90        | 2.38        | 49.55        |
| No weather | <b>Triline Transformer 4L</b> | <b>15.76</b> | <b>2.31</b> | <b>49.46</b> |
| Weather    | Direct MLP                    | 17.66        | 4.66        | 51.39        |
| Weather    | Triline LSTM 4L               | 16.06        | 2.39        | 51.14        |
| Weather    | <b>Triline Transformer 4L</b> | <b>15.95</b> | <b>2.28</b> | <b>45.46</b> |

We first evaluate the models on the next-day GPS prediction task, where the goal is to predict the bird’s location on the following day from the previous trajectory window. The proposed Triline Transformer 4L achieves the strongest overall performance among the compared models. In no-weather setting at  $k = 30$ , it obtains a mean error of 15.76 km and a median error of 2.31 km (Table 3), outperforming both the Direct MLP and Triline LSTM baselines. This result shows that the proposed model is effective not only in reducing average prediction error, but also in improving typical daily predictions.

The improvement over Direct MLP supports the benefit of structured movement prediction. Direct MLP predicts the next latitude and longitude directly from a flattened input window, so it treats the task mainly as coordinate regression. This is difficult for migration trajectories because the same recent location can lead to very different next-day outcomes depending on whether the bird remains at a stopover site or begins a migration movement. In contrast, the triline formulation separates the prediction into movement-related components before reconstructing the GPS location. This gives the model a more behavior-aligned representation of the task. Quantitatively, Direct MLP has a mean error of 18.43 km and a median error of 5.26 km, while Triline Transformer reduces these to 15.76 km and 2.31 km. The large reduction in median error suggests that the structured formulation is especially helpful for common daily movement cases.

The comparison with Triline LSTM isolates the value of attention-based temporal modeling. Unlike Direct MLP, Triline LSTM uses the same movement-based prediction formulation and also models

temporal context across the input window. Its stronger performance compared with Direct MLP indicates that both movement decomposition and sequential context modeling are useful. However, Triline Transformer still performs best overall, suggesting that self-attention provides an additional advantage. The key difference is how the two models use the history: LSTM summarizes the past window through a sequential hidden state, while Transformer can directly compare all days in the context window and attend to the days most relevant to the next movement. This is useful for migration trajectories, where the next-day location may depend on patterns such as recent stopover duration, changes in heading, or the onset of a sustained southbound movement.

Overall, these results show that the Triline Transformer 4L benefits from both structured movement prediction and attention-based temporal modeling. Compared with direct GPS regression and recurrent sequence modeling, it provides a more behavior-aligned and stable approach for next-day bird migration forecasting.

## 6.2 Architecture and Context Trends

We next analyze whether the Triline Transformer benefits from longer trajectory context in next-day migration forecasting. In the no-weather setting, the Triline Transformer 4L achieves mean errors of 17.38 km, 17.62 km, and 15.76 km for  $k = 7$ ,  $k = 14$ , and  $k = 30$ , respectively (Figure 2). The best performance occurs at  $k = 30$ , where the mean error is reduced by 1.62 km compared with  $k = 7$ , corresponding to a relative improvement of about 9.3%. This suggests that next-day migration prediction depends on more than immediate recent motion. A short window can capture local displacement, but a longer window can better represent whether the bird is stopping, restarting movement, or following a sustained southbound trajectory.

The non-monotonic trend between  $k = 7$  and  $k = 14$  shows that adding more history is not automatically beneficial. The  $k = 14$  model performs slightly worse than  $k = 7$ , suggesting that an intermediate window may add older trajectory variation without fully capturing a coherent migration phase. In contrast,  $k = 30$  corresponds roughly to a one-month segment and appears to better match the temporal scale of seasonal migration behavior in this dataset. This interpretation is consistent with ecological movement literature, where animal movement is often described as a sequence of steps shaped by behavioral state and environmental constraints (6), and where atmospheric conditions can produce delays, detours, and favorable movement corridors during migration (2). For the Transformer, the longer window is useful because self-attention can select relevant days from the full context, rather than relying only on the most recent observations.

Overall, the context experiment shows that the Triline Transformer 4L benefits most from a 30-day trajectory window. This does not imply that arbitrarily longer windows would continue to improve performance; rather,  $k = 30$  appears to provide a useful temporal scale for capturing migration-phase information in our data.

## 6.3 Weather Ablation

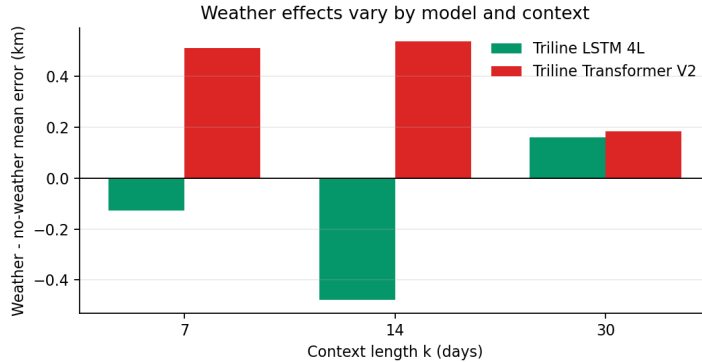


Figure 3: Weather-minus-no-weather mean error for triline models. Negative values indicate that adding weather reduces mean error.

Table 4: Weather ablation for triline models. Delta is weather minus no-weather mean error.

| Model                  | $k$ | No-weather mean | Weather mean | Delta |
|------------------------|-----|-----------------|--------------|-------|
| Triline LSTM 4L        | 7   | 17.94           | 17.82        | -0.13 |
| Triline LSTM 4L        | 14  | 18.21           | 17.73        | -0.48 |
| Triline LSTM 4L        | 30  | 15.90           | 16.06        | +0.16 |
| Triline Transformer 4L | 7   | 17.38           | 17.89        | +0.51 |
| Triline Transformer 4L | 14  | 17.62           | 18.16        | +0.54 |
| Triline Transformer 4L | 30  | 15.76           | 15.95        | +0.18 |

We further evaluate whether ERA5 weather variables improve next-day prediction for the triline models. As shown in Figure 3 and Table 4, the effect of weather is not uniformly positive and depends on both architecture and context length. For Triline LSTM, weather slightly reduces mean error at shorter contexts, improving by 0.13 km at  $k = 7$  and 0.48 km at  $k = 14$ , but the benefit disappears at  $k = 30$ . For the main Triline Transformer 4L, adding weather increases mean error at all context lengths: +0.51 km at  $k = 7$ , +0.54 km at  $k = 14$ , and +0.18 km at  $k = 30$ . Thus, raw ERA5 weather features do not consistently reduce average next-day GPS error for the Transformer.

For the Triline Transformer, the context trend is still informative. The weather penalty is much smaller at  $k = 30$  than at  $k = 7$  or  $k = 14$ , suggesting that longer trajectory context helps the model align environmental information with the bird’s movement state. In other words, weather variables may be difficult to use when the model only observes a short motion history, but become less disruptive when the model has enough context to infer whether the bird is stopping, restarting, or continuing southward.

Mean error also does not capture all weather-related effects. This is important because rare large errors are common in migration forecasting and can dominate practical path-level reliability. Although adding weather does not improve the Triline Transformer mean error, the  $k = 30$  P90 error decreases from 49.46 km to 45.46 km (Table 3). This suggests that weather may help on some harder prediction cases, even if the benefit is not large or consistent enough to reduce the overall average error.

These results suggest that the current weather representation is useful but not fully aligned with individual endpoint prediction. Unlike aggregate migration-flow forecasting, individual next-day GPS prediction is affected by bird-specific route choice, stopover behavior, and departure timing, which can weaken the apparent weather signal. In addition, our ERA5 variables are aggregated daily and matched to the bird’s daily location, which may be too coarse to capture departure-time conditions or weather along the future route. In short, trajectory history remains the dominant signal for mean error, while weather provides weaker and more case-dependent information.

#### 6.4 Autoregressive Rollout

Next-day accuracy is necessary but not sufficient for full-path forecasting. We therefore evaluate autoregressive rollouts on 27 valid held-out paths, using at least 60 observed days before rolling forward. These rollouts repeatedly feed predictions back into the model context, so small one-step biases can grow into large path-level errors.

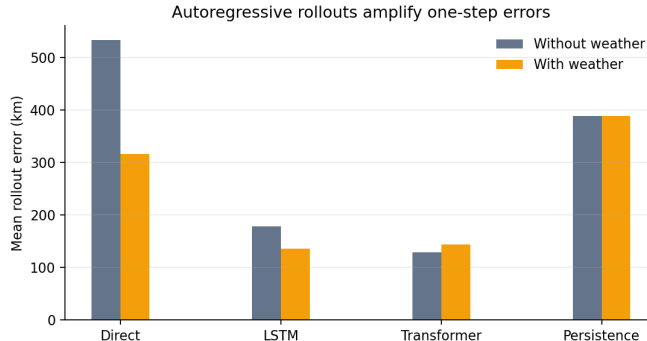


Figure 4: Mean autoregressive rollout error over 27 held-out paths. Lower is better.



Table 5: Autoregressive rollout errors over 27 valid held-out paths.

| Condition           | Direct MLP | LSTM          | Transformer   | Persistence |
|---------------------|------------|---------------|---------------|-------------|
| No weather mean km  | 534.17     | 178.37        | <b>129.55</b> | 388.62      |
| Weather mean km     | 316.03     | <b>136.61</b> | 144.11        | 388.62      |
| No weather final km | 877.29     | 273.35        | <b>178.28</b> | –           |
| Weather final km    | 480.35     | 180.47        | <b>166.68</b> | –           |

As shown in Figure 4 and Table 5, rollout errors are much larger than next-day errors for all models: while next-day mean errors are on the order of tens of kilometers, rollout mean errors increase to the hundred-kilometer scale. This confirms that trajectory forecasting requires not only accurate local prediction, but also stability in preserving the global migration direction over multiple future steps.

The Triline Transformer is the strongest model in the no-weather rollout setting. It achieves the lowest mean rollout error of 129.55 km, compared with 178.37 km for LSTM, 534.17 km for the direct model, and 388.62 km for persistence. It also gives the lowest no-weather final endpoint error of 178.28 km. These results show that the triline movement formulation and attention-based temporal modeling improve path-level stability. Direct coordinate regression is especially fragile under rollout because coordinate errors are recursively reused, while persistence fails because it cannot represent sustained southbound migration.

Weather has a mixed effect in rollout, which is consistent with the next-day weather ablation. For the direct and LSTM models, adding weather reduces mean rollout error. For the Triline Transformer, weather increases mean rollout error from 129.55 km to 144.11 km, but improves final endpoint error from 178.28 km to 166.68 km. This suggests that raw daily weather features do not uniformly improve the intermediate predicted path, but may still provide useful information for the overall endpoint. In other words, weather signal appears helpful in some rollout metrics, but remains noisy and model-dependent.

Overall, the rollout experiment shows that the Triline Transformer better preserves long-horizon migration structure than direct regression or persistence, but error accumulation remains a central challenge for full trajectory forecasting.

## 7 Conclusion

We study individual-level southbound bird migration trajectory forecasting from recent GPS history and ERA5 weather covariates. Our main model is a Triline Transformer, which decomposes next-day movement into fly/no-fly, distance, and heading before reconstructing the next GPS location. We evaluate this structured formulation against direct coordinate regression and recurrent triline baselines under next-day prediction and autoregressive rollout.

Our results show that the triline formulation provides a more behavior-aligned way to model migration movement. The Triline Transformer 4L performs strongly across the main next-day prediction experiments, and its advantage over Direct MLP supports the value of predicting movement primitives rather than coordinates directly. Its comparison with Triline LSTM further suggests that self-attention helps use temporal context more flexibly. The context experiment shows that  $k = 30$  is most effective for the Triline Transformer, indicating that one-month trajectory history better captures migration-phase information such as stopover behavior and sustained southbound movement.

Weather features provide useful but model-dependent signal. Raw daily ERA5 variables do not consistently reduce mean error for the Triline Transformer, but they reduce some tail-error metrics and remain informative in rollout analysis. The main limitations are coarse daily weather matching, individual-level behavioral variability, and error accumulation during autoregressive rollout. Future work should reduce two sources of mismatch in the current pipeline. On the input side, weather matching should use sub-daily conditions near the estimated departure time and along the likely flight route, rather than relying only on daily averaged weather at the bird’s daily location. On the output side, rollout-aware training could make the model more robust when its own predictions are reused during long-horizon forecasting.

## 8 Code and GitHub

The code and experiment artifacts are available at: <https://github.com/cjyaddone/ECE-228-Project>.

Table 6: Artifacts used to reproduce the final report.

| Item                    | Location or value                                       |
|-------------------------|---------------------------------------------------------|
| Dataset                 | data/combined_southbound_paths_with_weather_matched.csv |
| Final experiment script | model_files/run_final_path_experiment.py                |
| Rollout/GIF script      | model_files/run_current_rollout_gifs.py                 |
| Model definitions       | model_files/model.py                                    |
| Result summaries        | with_weather/, without_weather/, rollout/               |
| Random seed             | 42                                                      |
| Hardware target         | CUDA GPU when available                                 |

## 9 Team Member Contribution

Both team members contributed to all major parts of the project, including problem formulation, literature review, data preprocessing, feature construction, model implementation, experiment design, result analysis, presentation preparation, and final report writing. Each member participated in discussions throughout the project and contributed to reviewing and editing the final submission.

## 10 Appendix

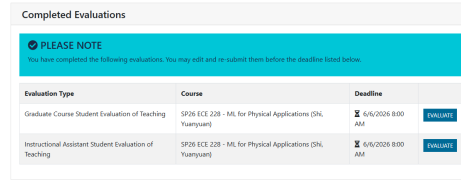


Figure 5: Screenshot of evaluation submission (Jiayi Chen)

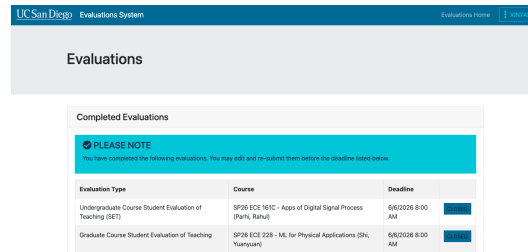


Figure 6: Screenshot of evaluation submission (Xinyan Cai)

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